The Law of Graphic Equalisation — Born Recovery

An Axiom-by-Axiom Note

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DRAFT — draft-2026-09-02.1 r1 (2 September 2026). This document is under active revision and is circulated for comment, not as a finished result. It is a companion to *The Law of Graphic Equalisation — Theory* (referred to below as Graphic Equalisation) and to *The Law of Graphic Equalisation — Physics Grounding* (the physics companion), and inherits those papers' definitions, epistemic boundary, and level-of-claims. It is an *attempt*, rated step by step, and not a peer-reviewed result. Sections marked as falsifiability commitments are the intended points of attack.

Abstract

This note checks, one axiom at a time, whether the change-substrate of Graphic Equalisation supplies the operational axioms from which the informational reconstructions of quantum theory derive the complex Hilbert space and the Born rule. It is self-contained; the papers carry the surrounding framework. The honest verb throughout is *recover*, not *derive*: the substrate reproduces the known rule within its own structures, given its admitted primitives and a contingent, self-located selection, rather than compelling it from necessity. Each step is rated solid, plausible or gap, and the residuals are named rather than hidden. Two findings carry the note. First, the substrate instantiates the operational-probabilistic core — convex states, affine effects, a transformation monoid with reversibles, composition, normalised probabilities — from the classical total-probability backbone and not from any quantum assumption. Second, two independent reconstructions have two *different* most-quantum axioms, and the substrate supplies each natively from a different one of its own principles: Hardy's continuity of reversible transformations from free continuation, and the purification of Chiribella, D'Ariano and Perinotti from entanglement as a superframe dimension. Complex over real is then not a gap to be closed by a forcing argument but the signature of a contingent selection, self-located exactly as the number of spatial dimensions is. The only claim made here is the *arrangement*; each component is individually known, and the uniqueness is supplied by the cited theorems.

1 The substrate, in one paragraph

Reality is modelled as typed graphs plus *admissibility rules* — a grammar of which edges may exist. A *frame* is a subgraph that runs; its *state* is an internal Narrative, the standing structure it carries forward. Where a frame instantiates a direction — optional in general, present in this setting — that Narrative has a magnitude and a *phase* that is position on a *closure cycle*, a loop that a realisation must close, so amplitudes are complex, $r\,e^{i\theta}$. A *realisation* is a collapse: a local write that picks one outcome from the admissible set, with a weight. Realisations are strictly ordered; there is no in-frame concurrency. Nothing is privileged: there is no god's-eye frame, and no privileged phase origin or measurement basis. When no rule fires, a frame's Narrative simply carries forward — *free continuation*, the reversible default. *Free* here means

information-preserving, not absent: evaluating a rule is itself a write, for the plain computing reason that a read is an operation and it *returns data*, and in a substrate that is graph all the way down there is no return channel outside the structure, so the returned value materialises as structure. What it *costs* is set by the substrate that runs the rules and not universally — Landauer (Landauer, 1961) is a floor in the physical substrate, not the price in any — and a frame cannot see its own runner's charge, which is why continuation presents as free from inside. Nothing in the mapping below depends on the economics: what it uses is that free continuation is *injective*, hence reversible, which is a structural fact independent of what anything charges for it. The invariant speed c is the frame's own change-throughput unit, 1 by construction, and therefore not directly readable from inside, since any in-frame measurement is funded by the same throughput and returns 1 whatever the underlying rate. A frame reaches a *number* for c only indirectly, by measurement plus retention plus inference, which is both why the unit-reading carries no risk and why the cross-frame claim is empirical at all. Frames compose by *nesting*, a subframe in a superframe; a correlation invisible in each subframe is a dimension of the superframe.

2 The claim, honestly sized

The substrate *recovers* the Born rule — reproduces it within its own structures, given its admitted primitives and a contingent, self-located selection — rather than *deriving* it as a forced necessity.

Remark 2.1 (Recovery is the honest verb, and the right one). Derivation per se would compel Born from primitives. The framework forces almost nothing, and that is the point of a graph substrate: what survives every attempted removal is a container relation across which rules run, and essentially everything else is contingent rule-content, as with the number of spatial dimensions or the gauge group. A contingent law can only be recovered given its rule-content, never derived as compelled. Some steps below are forced *given the premises* — the squared modulus, from no-privilege together with the additivity of decohered alternatives — while the premises themselves are admitted and contingent: complex, linear, self-similar.

Every distinctively-quantum ingredient is supplied by a substrate principle; the cited reconstruction theorems supply the uniqueness (Gleason, 1957; Hardy, 2001; Masanes and Müller, 2011; Chiribella et al., 2011). The only real claim is the *arrangement* — each component is individually known. That sizing is not modesty about the result but a statement of what kind of result it is, and it governs every rating below.

3 Level 0: the substrate is an operational-probabilistic theory

Before any reconstruction axiom applies, the substrate must be an operational-probabilistic theory (OPT): states in a convex set, effects, transformations, and a composition rule. That framework is a *precondition* for either axiom set, so the mapping has two levels, and this is the lower one.

The key that makes it tractable is seeing what the OPT convex structure *is*. The convexity of the state space and the affinity of effects are the *classical total-probability backbone* — total probability over ordered branchings — distinct from, and prior to, the quantum superposition linearity that Section 4 worries about. Every probabilistic theory has this backbone. The substrate has it by construction, because realisations carry weights, compose in order, and depend only on the present.

Proposition 3.1 (The substrate instantiates the operational-probabilistic core)**.** *Convex states, affine effects, a transformation monoid with reversibles, composition, and normalised probabilities are each supplied by the substrate's weighted, ordered realisations together with its history-optional,*

present-structure-acts first law. The instantiation rests on the classical total-probability backbone and on no quantum assumption.

Ingredient by ingredient, with the substrate structure that supplies each:

OPT requirement	**Supplied by**	**Rating**
Systems	Frame types: a subgraph together with an outcome set.	solid
States are well-defined equivalence classes	Statistics depend only on the current Narrative — history is optional and present structure acts, so there is no preparation-memory and operational equivalence holds.	solid
Convex state space	Mixing two preparations by a weighted coin-realisation: realise a λ-weighted coin, prepare P on one branch and Q on the other.	solid
Affine effects	Total probability over the coin–prepare–measure branching, $e(\lambda P + (1-\lambda)Q) = \lambda e(P) + (1-\lambda)e(Q)$.	solid
Transformations (monoid, with reversibles)	Rules compose, in order; free continuation is the identity, information-preserving and hence reversible.	solid
Composition	Subframe nesting; a joint preparation lives over the product outcome set. Existence is Level 0; the complex-specific tomography is the contingent layer of Section 5.	solid
Probabilities	Realisation weights in $[0, 1]$, normalised over a complete measurement.	solid

Remark 3.2 (The affinity is not the superposition linearity)*.* The affinity of effects is the law of total probability applied to ordered realisations, and nothing about it is quantum. It is worth marking because the two linearities are easy to run together, and the argument would be circular if they were: Level 0 earns the classical backbone, and Section 4 must earn the superposition linearity separately, on other grounds.

What remains at this level is hygiene rather than core: a finite tomographic dimension, in the sense that a state is fixed by finitely many fiducial effects — plausible for finite frames by the aperture, unproven in general; topological closure and compactness of the convex state space; the full composition axioms, associativity of sequential and parallel composition and the interchange law, which should follow from the ordering and nesting of realisations but each need checking; and the no-restriction and effect-realisability niceties. None of these is the "does it convex-linearise at all" worry that Level 0 originally named. That worry is resolved; these are carried forward in Section 7.

4 The distinctively-quantum axioms, mapped

Two independent reconstructions are used. Hardy (2001) gives five axioms, of which classical probability theory satisfies the first four and fails the fifth. Chiribella et al. (2011) — Chiribella, D'Ariano and Perinotti, CDP below — give six principles that single out quantum theory over the real, the quaternionic and the classical. They have *different* most-quantum axioms, which is what makes the pair worth running rather than either alone.

Proposition 4.1 (Each reconstruction's crux axiom is supplied natively, from a different substrate principle)**.** *Hardy's quantum-selecting axiom is the continuity of reversible transformations between pure states; the substrate supplies it from free continuation, which is information-preserving and hence reversible, and whose phase is a continuous $U(1)$, so it moves continuously between pure states. CDP's characteristically quantum principle is purification; the substrate*

supplies it from entanglement as a larger-frame dimension, a mixed subframe state being the restriction of a pure superframe state. Neither is borrowed from the other, and neither was introduced for this purpose.

That two separate reconstructions' crux axioms are both already-present substrate structure, from two different principles, is the substantive positive finding of this note. The remaining correspondences are weaker and are marked as such:

Axiom	**Substrate structure**	**Rating**
Continuity of reversible transformations (Hardy; classical theory fails exactly this)	Free continuation: information-preserving, hence reversible, with a continuous cycle-phase.	solid
Purification (CDP; fails classically)	Entanglement as a superframe dimension: a mixed subframe state is the restriction of a pure superframe state.	solid
Phase- and context-independence	No privileged phase origin and no privileged frame: probability is invariant under global phase and the same across measurement bases.	earned
Probability	Weighted realisations; frequencies over an ensemble of collapses from one possibility structure.	plausible
Simplicity and ideal compression	Minimum description length as a native commitment (Rissanen, 1978).	plausible
Subspaces	Subframe-as-frame: restriction to a subset of distinctions behaves as a smaller frame.	plausible
Perfect distinguishability	Distinctions are first-class; a state that is not completely mixed carries one.	plausible
Pure conditioning	Collapse of a pure Narrative yields a definite realised outcome.	plausible
Local tomography (Hardy's composite axiom; CDP's local distinguishability)	Subframe composition over the product outcome set — *contingent*, and the subject of Section 5.	contingent

The plausible ratings are genre-matches rather than statement-matches: the substrate has a commitment of the right kind, and that the specific axiom is the one the substrate enforces is not shown. Marking them so is the point of the table; a reader who upgrades one of them is disagreeing with a rating, which is the intended form of the disagreement.

Proposition 4.2 (The squared modulus, given complex linear amplitudes and no privileged phase)**.** *Decohered alternatives are classically exclusive, so their probabilities add. Averaging the combined modulus over the relative phase,*

$$\left\langle f(|a\,e^{i\theta}+b|)\right\rangle_{\theta} = f(|a|)+f(|b|),$$

since the cross term $2\,\mathrm{Re}(a^{*}b\,e^{i\theta})$ *averages to zero. The square,* $f(x)\propto x^{2}$*, satisfies this and the obvious alternatives do not: a quartic overshoots by the surviving* $\langle\cos^{2}\rangle=\frac{1}{2}$ *term, and the linear undershoots. This* selects *the square among the natural candidates; it does not prove it unique.*

The uniqueness is exactly what Gleason supplies (Gleason, 1957): over all measurement contexts, $|\cdot|^{2}$ is the only such measure in dimension at least three. What the framework supplies to Gleason are his premises — a vector-valued state, from the linearity step, and a context-independent measure, from the absence of a privileged rule set — and Gleason supplies what the framework does not.

Remark 4.3 (Two supporting identifications). The *inner product* is the cross-frame projection: one frame's Narrative read in another's basis, so it is not imported structure but the framework's own cross-frame reading. *Unitarity* is that a free continuation destroys no distinction — an

information-preserving write rather than the absence of one, which is the right shape, since unitary evolution plainly does change the state and is reversible for exactly that reason. Both are identifications rather than derivations, and the rigorous step from "uncoupled" to full additivity and norm-preservation is the standing residual of Section 7.

5 Complex over real: contingent, not forced

Local tomography is the axiom that selects complex Hilbert space over real, and it is where a forcing argument would have to sit. Writing K for Hardy's fiducial degrees of freedom including normalisation, for two two-state systems:

Theory	K **single**	K **composite**	$K_A \cdot K_B$	**Local tomography**
Classical	2	4	4	holds (but fails continuity)
Real (rebit)	3	10	9	**fails** ($10 > 9$)
Complex (qubit)	4	16	16	**holds**

A closure-phase frame is *consistent with* complex amplitudes at one system — the Bloch sphere, $K = 4$, rather than the real disk's $K = 3$ — though a phase-less frame is equally constructible and the substrate requires neither. Read that way, a superframe, being a frame, is a complex closure-phase Narrative over the product outcome set, giving $K_{AB} = 16$: local-tomographic, complex.

Proposition 5.1 (Complex over real is a self-located selection, not a forced consequence)**.** *The framework forces almost nothing. Complex-over-real is contingent rule-content, exactly as the number of spatial dimensions is. Local tomography is therefore not a gap to be closed by a forcing argument; it is the signature of a contingent selection. A real-amplitude composition, $K_{AB} = 10$, is another admissible configuration. Our universe is the complex one, self-located.*

Remark 5.2 (Why the forcing reading was a misframing)*.* Treating "does the substrate *force* the complex composite?" as the open question was an error of the framework's own kind, not merely an unproven step. Seeking to force complex over real would contradict the framework's stance that it never forces contingent physics. The reading that survives is narrower and does real work: the admissible complex configuration yields $K_{AB} = 16$ and the admissible real one yields 10; a real composite would be a superframe that is not a closure-phase frame like its subframes, so real amplitudes cost self-similarity. That gives a direction and a reason without giving a compulsion, and it is the same status every other contingent constant in this programme has.

Given the contingent choice, the Born rule and the Hilbert structure follow via the mapping and the cited reconstructions — which is how the framework treats all physics: shapes follow from contingent rule-content, the rule-content itself never forced. Among the post-2011 reconstructions, Masanes and Müller (2011) obtain finite-dimensional quantum theory from five physical requirements on states and transformations, and their formulation is the one this note would most like tested against, since its requirements are stated on single systems plus a composition rule — which is exactly the seam where the substrate's answer turns contingent rather than forced. Höhn (2017) and Höhn and Wever (2017) reach qubit quantum theory from rules on an observer's acquisition of information, and are the nearest in shape to the substrate's own machinery; worth noting on the other side of the ledger is that those rules yield the correlation structure of qubits and rebits alike, so they do not by themselves select complex over real, which is independent support for reading the selection as contingent. Selby et al. (2021) state every postulate diagrammatically, with symmetric purification central, which is the nearest in *formal register*: a compositional, diagrammatic axiom set is written in the same currency as a typed graph substrate with subframe nesting, so the mapping attempted here is a translation between two graphical calculi rather than between a graph and a convex body.

That is a reason to think the mapping can be checked, not evidence that it holds. Müller (2021) surveys the programme as a whole.

6 What this reconciles

On one substrate, quantum theory and relativity are not rivals: *collapse* is an in-frame event, a local write, discrete; *relativity* is a cross-frame map, continuous; and they are joined at c. Special and general relativity reconcile as one change-budget spent two ways, with the invariance of c a definitional consequence given the substrate's own units — the unit-reading itself carries no risk, and the falsifiable content is the cross-frame claim that aperture self-calibrates, so that two frames each obtaining c by the indirect route cannot be exhibited reaching different values for one signal. The quantum–relativity reconciliation is *structural* — one substrate, the event/map seam — with Born *recovered*.

This is emphatically **not** a quantum theory of gravity: no field is quantised, no metric derived, no dimensionless constant fixed. What is reconciled is the *conflict*, at the level of shape. A rival recovery from a comparably admissibility-flavoured base reaches a different answer from the same kind of premises — constructor theory (Deutsch, 2013), in which probability is not fundamental at all and what look like probabilities emerge from the unpredictability of measurement on superinformation media (Marletto, 2016) — and the recovery here is exposed to it rather than reinforced by it. The frame-relative reading of collapse on which all of this rests has its nearest ancestor in relational quantum mechanics (Rovelli, 1996).

7 Residuals

What is open, unaddressed, and named:

1. **Level 0 hygiene.** Whether "present structure acts" genuinely delivers operational equivalence — states as equivalence classes of preparations with identical statistics — and whether the full OPT composition axioms (associativity of sequential and parallel composition, the interchange law) hold under the ordering and nesting of realisations. With them, a finite tomographic dimension, topological closure and compactness of the state space, and no-restriction and effect-realisability. These are hygiene rather than core, and plausible for finite frames; none is proven here.

2. **The linearity step.** The rigorous passage from "uncoupled free continuation" to full additivity and norm-preservation. The mechanism motivates linearity; it does not prove it, and the linearity of the imposed supergraph map is not handed over for free. This is the one posit the recovery rests on.

3. **The squaring argument.** Whether the phase-average selection is tight enough on its own, or whether the full Gleason content is needed to reach uniqueness. As stated in Proposition 4.2 it selects rather than proves, and Gleason is cited for the rest.

4. **Composition.** Under the natural reading the composite is local-tomographic and complex. Whether that reading is forced, permitted, or wrong is not settled here. The formulation of Masanes and Müller (2011) is the one this seam should be tested against, its requirements being stated on single systems plus a composition rule; that is exactly where the substrate's answer turns contingent rather than forced, and where the mapping is least confident that it is doing honest work.

5. **The plausible ratings.** Probability, simplicity and compression, subspaces, distinguishability and pure conditioning are genre-matches to native commitments, not statement-matches. Each would need tightening before the cited theorems could be applied without qualification.

6. **In-frame energy as a count.** The identification on which the spectrum floor of the physics companion rests is clean in the occupation-number picture and less clean for interacting systems, where the count is not sharp and the physical states are dressed. It is not established here for that case.

8 Falsifiability commitments

Each commitment below is marked [SHAPE] or [DETAIL]. A [DETAIL] failing is a revision: the account survives with that mechanism replaced. A [SHAPE] failing costs *range* rather than a mechanism, because what fails is something the substrate is claimed to require or to permit — so the account does not hold where it claimed to, which is the more serious of the two and is still a boundary rather than an annihilation.

1. [SHAPE] **The operational-probabilistic instantiation.** Fails if the substrate does not instantiate an OPT — if convex states, affine effects, a transformation monoid with reversibles, composition and normalised probabilities cannot all be had from weighted ordered realisations and the history-optional first law — or if the hygiene conditions of Section 7 cannot be met for the frames this note is about. Everything below is conditional on this, so it fails with it.

2. [DETAIL] **Linearity from free continuation.** Fails if the step from uncoupled independent continuation to full additivity and norm-preservation cannot be made rigorous — in which case the vector structure is a posit rather than a recovery, and Gleason's first premise is imported rather than supplied.

3. [SHAPE] **The axiom mapping.** Fails if it does not go through for a named axiom: in particular if free continuation is not Hardy's continuous reversible transformation between pure states, or if entanglement as a superframe dimension is not CDP's purification. Failure at either of those two is worse than failure elsewhere, since they are the axioms that separate quantum from classical and are what Proposition 4.1 claims.

4. [DETAIL] **The squared modulus.** Fails if, given complex linearly superposing amplitudes and no privileged phase, the probability is not the squared modulus; and the selection argument of Proposition 4.2 fails separately if another natural candidate satisfies the phase-averaged additivity, leaving the whole weight on Gleason.

5. [SHAPE] **Complex over real is contingent.** Fails if the framework must *force* complex over real rather than admit both and self-locate — that is, if a substrate that admits a real-amplitude composition is thereby shown to be the wrong substrate, rather than one whose selection among admissible configurations is a matter of where the observer is.

6. [DETAIL] **Recovery, not derivation.** Fails as an honest sizing if any step above is read as compelling the Born rule from primitives. The claim is the arrangement, and it is refused if the arrangement is presented as a necessity.

Scope note (not a falsifier). This note is shape-level and not quantitative: it fixes no dimensionless constant, derives no field equation, and offers no theory of quantum gravity.

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